

Problems on k-NN algorithm:

- 2 types $\rightarrow$ Discrete-valued target function - (1)
 $\rightarrow$ Continuous-valued target function - (2)

Approximation function:

$$(1) \hat{f}(x_q) = \underset{v \in V}{\operatorname{argmax}} \sum_{i=1}^k \delta(v, f(x_i)) \text{ where, } \delta = \sqrt{(v - x_i)^2}$$

$$(2) \hat{f}(x_q) = \frac{\sum_{i=1}^k f(x_i)}{K}$$

Discrete-Value Problems:

1)

Sl.no	Height(cm)	Weight(kg)	Target(v)	Distance	Neighbors
1	150	50	M	8.06	N_4
2	155	55	M	2.24	N_1
3	160	60	L	6.71	N_3
4	161	59	L	6.40	N_2
5	158	65	L	11.05	N_5
Test	157	54	L	?	$K=3 \text{ or } 4$

Either will be given or not, can be calculated

Distance,

$$\delta = \sqrt{(150 - 157)^2 + (50 - 54)^2} = \sqrt{49 + 16} = 8.06$$

Training, $K=4$

$$\begin{aligned} \hat{f}(x_q) &= \delta(M, f(x_2)) + \delta(M, f(x_3)) + \delta(M, f(x_4)) + \delta(M, f(x_5)) \\ &= \delta(M, M) + \delta(M, L) + \delta(M, L) + \delta(M, L) \\ &= 1 + 0 + 0 + 0 = 1 \end{aligned}$$

Testing,

$$\begin{aligned} \hat{f}(x_q) &= \delta(L, f(x_2)) + \delta(L, f(x_3)) + \delta(L, f(x_4)) + \delta(L, f(x_5)) \\ &= 0 + 1 + 1 + 1 = 3 \end{aligned}$$

Training, $K=3$

$$\begin{aligned} \hat{f}(x_q) &= \delta(M, f(x_2)) + \delta(M, f(x_3)) + \delta(M, f(x_4)) \\ &= \delta(M, M) + \delta(M, L) + \delta(M, L) \\ &= 1 + 0 + 0 = 1 \end{aligned}$$

Testing,

$$\begin{aligned} \hat{f}(x_q) &= \delta(L, M) + \delta(L, L) + \delta(L, L) \\ &= 0 + 1 + 1 = 2 \end{aligned}$$

Continuous Value Problems:

Sl. No	Height (cm)	Weight (kg)	Target	Distance	Neighbors
1	150	50	1.5	8.06	N_4
2	155	55	1.2	2.24	N_1
3	160	60	1.8	6.71	N_3
4	161	59	2.1	6.40	N_2
5	158	65	1.7	11.05	
Test	157	54	1.65	?	$k=3$ or 4

$$\hat{f}(x_n) = \frac{1.2 + 2.1 + 1.8 + 1.5}{4} = 1.65$$

Problems on Naive Bayes Algorithm:

Color	Legs	Height	Smelly (?)	Species (T)
White	3	Short	Yes	M
Grey	2	Tall	No	M
Grey	3	Short	Yes	M
White	3	Short	Yes	M
Grey	2	Short	No	H
White	2	Tall	No	H
White	2	Tall	No	H
White	2	Short	Yes	H

Test:

$X = \{\text{Grey}, 2, \text{Tall}, \text{No}\}$

$$P(M) = \frac{4}{8} \quad P(H) = \frac{4}{8}$$

Color	M	H
White	$\frac{2}{4}$	$\frac{3}{4}$
Grey	$\frac{2}{4}$	$\frac{1}{4}$

Legs	M	H
3	$\frac{3}{4}$	$\frac{0}{4}$
2	$\frac{1}{4}$	$\frac{4}{4}$

Height	M	H
Short	$\frac{3}{4}$	$\frac{2}{4}$
Tall	$\frac{1}{4}$	$\frac{2}{4}$

Smelly (?)	M	H
Yes	$\frac{3}{4}$	$\frac{1}{4}$
No	$\frac{1}{4}$	$\frac{3}{4}$

$$P(M|x) = P(M) \times P(\text{Grey}|M) \times P(2|M) \times P(\text{Tall}|M) \times P(\text{No}|M)$$

$$= \frac{1}{2} \times \frac{1}{2} \times \frac{1}{4} \times \frac{1}{4} \times \frac{1}{4}$$

$$= 0.0039$$

$$P(H|x) = P(H) \times P(\text{Grey}|H) \times P(2|H) \times P(\text{Tall}|H) \times P(\text{No}|H)$$

$$= \frac{1}{2} \times \frac{1}{4} \times 1 \times \frac{1}{2} \times \frac{3}{4}$$

$$= 0.0468$$

$$N.B = \arg\max(P(H|x), P(M|x))$$

$$= 0.0468$$

Target = H

②

Weather	Road	Traffic	Engine	Accident (T)
Rain	Bad	High	No	Yes
Snow	Avg	Normal	Yes	Yes
Clear	Bad	Low	No	No
Clear	Good	Low	Yes	Yes
Snow	Avg	Normal	No	No
Rain	Good	Low	No	No
Rain	Bad	Normal	No	No
Snow	Bad	High	No	Yes

Test:

$$X = \{\text{Snow, Good, Low, No}\}$$

$$P(\text{Yes}) = \frac{4}{8}, \quad P(\text{No}) = \frac{4}{8}$$

Weather	Yes	No	Road	Yes	No	Traffic	Yes	No
Rain	$\frac{1}{4}$	$\frac{2}{4}$	Good	$\frac{1}{4}$	$\frac{1}{4}$	High	$\frac{2}{4}$	$\frac{1}{4}$
Snow	$\frac{2}{4}$	$\frac{1}{4}$	Bad	$\frac{2}{4}$	$\frac{2}{4}$	Low	$\frac{1}{4}$	$\frac{2}{4}$
Clear	$\frac{1}{4}$	$\frac{1}{4}$	Avg	$\frac{1}{4}$	$\frac{1}{4}$	Normal	$\frac{1}{4}$	$\frac{2}{4}$

Engine	Yes	No
Yes	$\frac{2}{4}$	$\frac{0}{4}$
No	$\frac{2}{4}$	$\frac{1}{4}$

$$P(\text{Yes}|x) = P(\text{Yes}) \times P(\text{Snow}|\text{Yes}) \times P(\text{Good}|\text{Yes})$$

$$\times P(\text{Low}|\text{Yes}) \times P(\text{No}|\text{Yes})$$

$$= \frac{1}{2} \times \frac{1}{2} \times \frac{1}{4} \times \frac{1}{4} \times \frac{1}{2} = 0.0078$$

$$P(\text{No}|x) = P(\text{No}) \times P(\text{Snow}|\text{No}) \times P(\text{Good}|\text{No})$$

$$\times P(\text{Low}|\text{No}) \times P(\text{No}|\text{No})$$

$$= \frac{1}{2} \times \frac{1}{4} \times \frac{1}{4} \times \frac{1}{2} \times 1 = 0.0156$$

$$N.B = \arg\max(P(\text{Yes}|x), P(\text{No}|x)) = 0.0156$$

Target = No